Daffodil International University

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CSE 124 || Data Structure Lab

NEXUS Rescue

Disaster Rescue Route Optimizer with Dynamic Hazard Propagation

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Problem Statement

Bangladesh is one of the world's most disaster-prone countries, facing annual floods, cyclones, and earthquakes. When disaster strikes, rescue teams must navigate through severely damaged infrastructure including flooded streets, collapsed bridges, and debris-blocked roads under extreme time pressure. Currently, rescue operations rely on manual decision-making and radio communication, which is slow and error-prone. There is no real-time, data-driven system that computes optimal rescue routes while accounting for dynamically changing hazard conditions. As a result, critical time is wasted and victims with limited survival windows may not receive help in time.

Objective

NEXUS Rescue aims to build an interactive web-based simulation that:

- Models a disaster-affected area as a weighted graph (nodes = locations, edges = roads).
- Dynamically removes or re-weights edges as hazards propagate in real time.
- Uses Dijkstra's Algorithm with a Priority Queue to compute the shortest safe rescue route.
- Assigns multiple rescue teams to victims without conflict using a greedy strategy.
- Provides an Admin Panel for authorized operators to control the map and a Public View for general awareness.
- Visualizes the entire routing process step-by-step on an interactive web map.

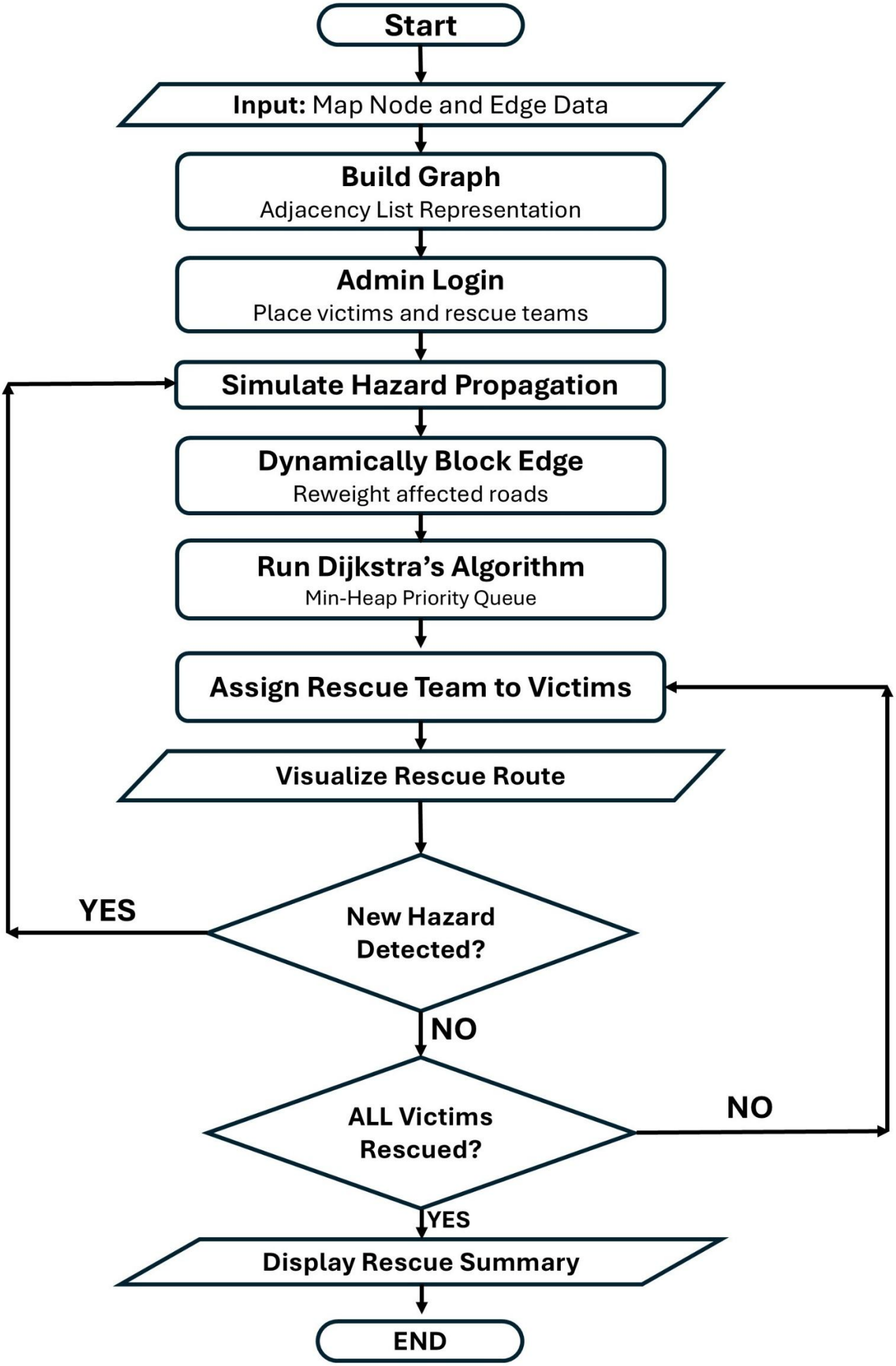
Software & Tools:

Category	Tool / Technology
Core Algorithm	C
Frontend	HTML, CSS, JavaScript
Version Control	Git & GitHub
IDE	VS Code / Code::Blocks

Data Structures Used:

- Graph (Adjacency List)
- Priority Queue
- Queue
- Stack
- Array

FlowChart:


Start
Input: Map Node and Edge Data
Build Graph
Adjacency List Representation
Admin Login
Place victims and rescue teams
Simulate Hazard Propagation
Dynamically Block Edge
Reweight affected roads
Run Dijkstra's Algorithm
Min-Heap Priority Queue
Assign Rescue Team to Victims
Visualize Rescue Route
New Hazard Detected?
YES
NO
ALL Victims Rescued?
NO
YES
Display Rescue Summary
END

Limitation

- **No Real Map Data:** A manually designed fictional map is used instead of live GPS or GIS data, as real map APIs require premium access.
- **Manual Hazard Input Only:** Disaster updates are entered by an Admin operator with no automated IoT sensor integration in this version.
- **Simplified Visualization:** Route animation is browser-based with no native mobile app or AR interface.
- **Basic Team Coordination:** A greedy assignment strategy is used and advanced optimization is out of scope.
- **No Live Disaster Feed:** Hazard scenarios are simulated and not pulled from real-time disaster databases.

Future Scope

- **IoT Sensor Integration:** Real flood and seismic sensors can feed live data directly, making hazard propagation fully automatic.
- **Drone Navigation:** The routing algorithm can guide autonomous rescue drones to deliver supplies or locate victims.
- **Mobile Application:** Rebuilt as an Android/iOS app with GPS tracking for field rescue workers.
- **Machine Learning Layer:** Past disaster data can train a model to predict road failures proactively.
- **National Emergency Network:** Scale to a country-wide rescue coordination platform, which is critical for Bangladesh's flood-prone regions.

Conclusion

NEXUS Rescue demonstrates how core data structures including graphs, priority queues, and BFS can be applied to a genuinely critical real-world problem. Disaster response is a domain where every second counts and an optimized routing system can directly save human lives. Through this project, we bridge theoretical data structure knowledge with practical emergency management while building a system that is meaningful, extensible, and technically rigorous.

References

- Bangladesh Disaster Management Bureau. www.ddm.gov.bd
- GeeksforGeeks. Dijkstra's Algorithm in C. www.geeksforgeeks.org
- Wikipedia.
- AI Chatbot (ChatGPT, Gemini)